

Petrographic Analysis of Apollo Sample 12004-11-r1

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1 Introduction

Paired PPL/XPL rotation series for Apollo lunar sample 12004-11-r1 was obtained initially from the Virtual Microscope. These petrographic images were subsequently aligned to ensure a consistent field of view by using Python. The rock sample 12004 was collected on (or near) the rim of Middle Crescent Crater during the Apollo 12 mission.

2 Phenocrysts

2.1 Dominant phase

The bulk of the section is composed of large anhedral to subhedral, equant grains that are colorless (or white) and non-pleochroic in PPL with moderate-to-high positive relief and no cleavage, instead showing the characteristic pervasive irregular/conchoidal fracturing. Under crossed polars these grains display high birefringence, with vivid upper-second- to third-order interference colors (magenta, purple, blue, yellow, green). Through the rotation series individual grains pass cleanly into and out of extinction at successive stage positions, consistent with the sharp extinction expected of olivine. The combination of colorless high relief, lack of cleavage, conchoidal fracture, and high-order interference colors is diagnostic of olivine.

2.2 Blocky brownish grains

Blocky grains with one to two cleavage traces and second-order interference colors, locally with weak brownish tints. Present in roughly equal proportion to olivine among the phenocrysts, with a comparable size range. Those are tentatively pyroxene.

3 Variolitic groundmass

The interstitial groundmass (60% of the rock) is built of plagioclase and pyroxene intergrown as sheaf-like, fan-shaped, sub-radiating bundles (varioles) that nucleate on common centers and on the margins of the larger crystals and splay outward from them. In PPL, the texture reads as feathery sprays of fine, dark, acicular-to-bladed crystals threading between the phenocrysts. Under XPL, the bundles carry low-to-moderate (first-order) interference colors and diagnostically show sweeping/undulatory extinction: as the stage rotates, each fan extinguishes progressively from one side to the other rather than all at once, because its constituent fibers fan across a range of

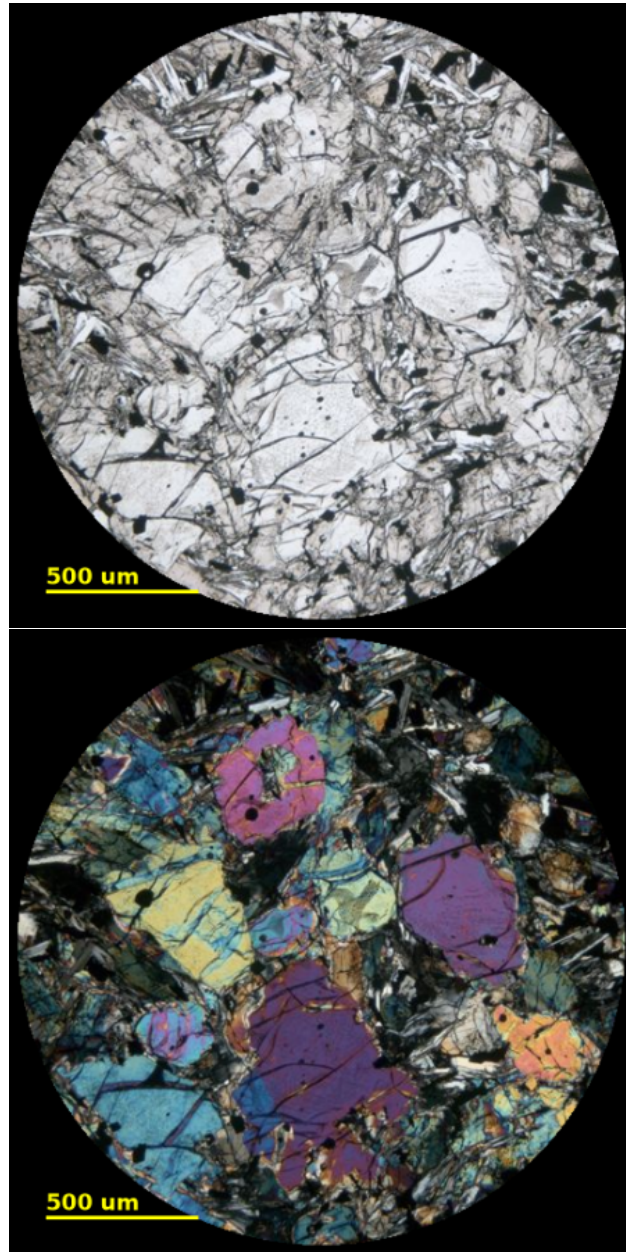


Figure 1: Sample 12004-11-r1 in PPL (top) and XPL (bottom). Field of view 2.0 mm; scale bar 500 μm . Equant olivine and blocky clinopyroxene phenocrysts (high-order interference colors in XPL) sit in a finer variolitic groundmass of plagioclase & pyroxene.

orientations. The plagioclase laths are thin and locally hollow/skeletal (intrafasciculate), consistent with rapid growth. Measured fan-to-fan, the varioles are 0.1–0.4 mm across; individual fibers are only 10–25 μm wide. Following Brett and coworkers, the texture is appropriately described as variolitic to sub-variolitic.

4 Opaque and accessory phases

Skeletal to lath-shaped opaque grains are intergrown within the variolitic bundles and along pyroxene margins. Most probably, equant, blocky chromite occurs as discrete grains and as inclusions, locally with brown ulvöspinel rims characteristic of this basalt type; and minor blebs of troilite and native Fe-metal occur in the groundmass and in association with the spinel grains. No magnetite and no hydrous phases are present.

5 Interpretation

The order-of-magnitude step from sub-millimeter olivine & clinopyroxene phenocrysts to 10–20 μm groundmass fibers records a two-stage cooling history typical of a lunar mare lava: slow early growth of olivine and pyroxene, then rapid undercooling that drove fascicular, variolitic crystallization of the residual plagioclase-bearing melt.

Glossary

Porphyritic texture An igneous texture characterized by large crystals (phenocrysts) embedded in a finer-grained groundmass, indicating two-stage cooling.

Variolitic texture A texture consisting of small radiating clusters of needle-like crystals (varioles) within a fine-grained or glassy matrix, typically formed by rapid cooling.

Phenocryst A relatively large crystal that formed early in the cooling history of a magma.

Relief Apparent prominence of a grain's outline and surface in PPL, reflecting the refractive-index contrast with its surroundings.

Pleochroism A change in absorption color as the stage is rotated under plane-polarized light.

Birefringence The difference between a mineral's principal refractive indices; it controls the order of interference colors seen under XPL.

Interference colors Colors seen under crossed polars, grouped by order (first, second, third...) according to retardation.

Extinction The stage orientation at which an anisotropic grain appears dark under XPL.

Groundmass The finer-grained matrix surrounding larger crystals in an igneous rock.

Acicular Needle-shaped.

Skeletal / hollow crystal A crystal whose faces are formed but whose interior is incompletely filled, produced by rapid growth.

Anhedral / subhedral Crystal outline with no / partial / complete development of crystal faces.

Conchoidal fracture Smooth, curved fracture surfaces not controlled by cleavage (characteristic of olivine).

Sweeping /undulatory extinction Extinction that migrates across a grain or fibre bundle as the stage rotates, rather than occurring uniformly; diagnostic of varioles here.

Modal proportion The volume percentage of a given phase in the rock.